IT 540: Introduction to Data Science
Data description for Week 5 assignment on Neural Nets.

The raw input data is contained in the file matmob.data.sample.Rdata OR for Python it's in matmob.data.sample.csv. For R users only: After you load that data into R, it will automatically be bound to the variable matmob.data.sample as a data frame.

The data frame has 100,000 observations of 16 variables, Each observation corresponds to a chess game taken from a larger collection of nearly 1.7 million games downloaded from the web. In addition, you should see row numbers on the left hand margin.

Each observation contains information on an expert-level or above (Elo 2000 or higher) chess game. Most metadata, e.g. names of the players, date and location of the game and so on, is contained in a separate file that you do not have.

In addition to some data that chess players will recognize, there are some additional variables that were computed from the raw game data. Of particular interest, these variables relate to the chess concepts of 'material' and 'mobility'.

*The purpose of this assignment is for you to predict the outcome (result) of a chess game by looking **only** at the mean game material and mobility variables for each of the two players, and the game length in what are called 'half moves'.*

Material: In chess, the pieces are typically regarded as having point values. A player's material at any given state is the sum of that player's piece values on the board. Although the object of the game does not depend on piece points in any way, they are considered useful knowledge to players during a game as a rough measure of advantage (or disadvantage) relative to the opponent, as pieces are captured and removed from the board. If one player has 3 more points than the opponent on the board, it is not uncommon for that opponent to resign. At the highest levels of play, a material advantage of 1 point is often considered significant, other things being equal (which they very often are not). While there is no universally accepted valuation of the pieces, perhaps the most common is shown in the following table:

Piece Name	Point Value
Pawn	1
Knight	3
Bishop	3
Rook	5
Queen	9

Kings cannot be captured, and since each player always has one King on the board, their point values can be ignored with respect to material advantage. At the beginning of the game, both players have a total value of 39 piece points, given the valuations in the table above.

Mobility: Players take turns to move, with White always playing the first move. At any given turn, the side to play may have a great many possible moves, or no moves at all, in which case the game is over. The number of legal moves at any given turn is referred to as that player's

mobility. As with material, an advantage in mobility is generally considered an asset. However in general, even when there are numerous legal moves for a player, only one or a small number of them is considered worthy of serious consideration. A 'full' move is considered to have taken place when both White and Black have taken their corresponding turns, and a 'half' move is when either White or Black moves.

Variable Descriptions:

w.mat.mean : A numeric variable consisting of the mean material for White over the course of an entire game. It is computed by summing up the total White material for all that player's moves and dividing by the number of White moves.

w.mat.sd: A numeric variable, the standard deviation of w.mat.mean.

b.mat.mean: A numeric variable, the game mean material for Black, computed analogously to w.mat.mean.

b.mat.sd: A numeric variable, the standard deviation of b.mat.mean.

w.mob.mean: A numeric variable, the mean game mobility for White, computed analogously to w.mat.mean.

w.mob.sd: A numeric variable, the standard deviation of w.mob.mean.

b.mob.mean: A numeric variable, the mean game mobility for Black.

b.mob.sd: A numeric variable, the standard deviation of b.mob.mean.

result: Factor variable with 3 levels, 0-1,1-0, and 1/2-1/2. In chess notation, '0-1' refers to a win by Black, '1-0' refers to a win by White and '1/2-1/2' designates a draw.

valid.games.row: An integer variable, where the number refers to a separate dataset row that includes additional metadata for each game.

w.mat.vector: Character variable that includes the comma-separated string representation of the material at each of White's turns.

b.mat.vector: Character variable that includes the comma-separated string representation of the material at each of Black's turns.

w.mob.vector: Character variable that includes the comma-separated string representation of the mobility (number of legal moves) for each of White's turns.

b.mob.vector: Character variable that includes the comma-separated string representation of the mobility (number of legal moves) for each of Black's turns.

pgn: Factor variable, where the string name of that factor shows the entire set of moves made by both players in algebraic chess notation for this game. 'pgn' refers to 'portable game notation', a standard representation for chess games.

half.moves: A numeric variable that consists of the number of half moves for the game, as described above. Thus, for example, a value of 71 would indicate a game where

each player made 35 moves, followed by one more move by White, at which point the game ended.